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Magnitude and timing of New Hebrides Arc rotation: Paleomagnetic evidence from Nendo, Solomon Islands

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Abstract. Paleomagnetic results from Nendo record a total rotation of the northwestern corner of the New Hebrides Arc of 52° ± 12° clockwise. Reversed magnetic polarity and a nannofossil assemblage constrain the age of the Nendo pole to 4.29 - 3.58 Ma. Paleomagnetic data from Malekula, farther south in the same arc, imply a rotation of only about 28° since the late Miocene, probably because these poles have been affected by rotation of Malekula in response to collision of the arc with the D'Entrecasteaux Zone aseismic ridge. A new late Miocene reconstruction of the New Hebrides and Vitiaz arcs based on the paleomagnetic evidence requires a rotation of 39° around the point of intersection of the two arcs. The same rotation restores both the northwestern and southeastern segments of the New Hebrides Arc, although rotation of the southeastern segment commenced earlier. A series of sequential movements of segments of the arc, occurring at different times but acting to keep the arc roughly collinear, may be driven by a mechanism which minimizes deformation of the subducted plate. Propagation of spreading into the northwestern corner of the North Fiji Basin after 4.3 Ma is implied by the paleomagnetic results, and is consistent with spreading anomaly patterns in the basin. When the New Hebrides and Vitiaz Arcs are restored to their position relative to the Lau Ridge at 10 Ma, the Fiji Platform can be accommodated after clockwise rotation of 140°, to a position in line with the New Hebrides Arc. The large rotation for Fiji is consistent with paleomagnetic evidence indicating at least 115° of rotation.

1. Introduction

One of the first applications of paleomagnetism to the tectonics of the southwest Pacific (Figure 1) was the analysis of the motion of the New Hebrides Arc [*Falvey*, 1978]. Clockwise rotation of the New Hebrides Arc from the late Miocene to the present was indicated by results from the Vanuatu sector of the arc, spanning the islands from Espiritu Santo to Tanna (Figure 2). *Falvey*'s [1978] study suggested that the rotation did not commence until 6 Ma and that the absolute magnitude of the rotation was about 28°. Clockwise rotation of the New Hebrides Arc is also implied by the triangular shape of the North Fiji Basin, which opened through back arc spreading following rifting of the New Hebrides Arc from the remnant Vitiaz Arc after arc reversal in the late Miocene [*Gill and Gorton*, 1973].

First impressions based on the geometry of the North Fiji Basin would suggest that the pole of rotation of the New Hebrides Arc, relative to a fixed Vitiaz Arc, should lie near the intersection of the two arcs [*Gill and Gorton*, 1973]. Nendo lies close to this "hinge." Anticlockwise rotation of the New Hebrides Arc through 39° relative to a fixed Vitiaz Arc around a pole at 9.5°S, 167°E produces a geometrically satisfactory reconstruction (Figure 3). However, the limit of about 28° placed on the absolute rotation of the New Hebrides Arc by the paleomagnetic results from Vanuatu required that the pole be farther from the hinge. Under this constraint, *Falvey* [1978] reconstructed Vitiaz, New Hebrides, Fiji, and Lau-Tonga bathymetry relative to a stationary Australian Plate at the time of arc reversal, which he took to coincide with the beginning of New Hebrides rotation at 6 Ma. To do so, he rotated the New Hebrides Arc through 24.4° anticlockwise around a pole at 6.1°N, 169.0°E and restored the Vitiaz Arc to its position on the Australian Plate prior to its transfer to the Pacific Plate (Figure 4). Many authors have since followed the Falvey reconstruction, in whole or part [e.g., *Carney and Macfarlane*, 1978; *Neef*, 1982; *Falvey and Greene*, 1988].

Other features of North Fiji Basin tectonics are not well satisfied by the Falvey reconstruction, however. In particular, this reconstruction places the Fiji Platform in front of the main axis of the Miocene Outer Melanesian Arc [*Falvey and Greene*, 1988] and rotates Fiji through only about 20°. Some authors have chosen to ignore these elements of the Falvey reconstruction, preferring a collinear alignment of the New Hebrides Arc and Fiji Platform and/or a greater rotation for Fiji [e.g., *Malahoff et al.*, 1982a, b].

Anticlockwise rotation of Fiji is a complementary feature of North Fiji Basin spreading. Early paleomagnetic studies of volcanics from Viti Levu by *Tarling* [1967] indicated anticlockwise rotation; later work by *James and Falvey* [1978], principally based on the Suva Marls, quantified this rotation to be approximately 21° since 4 Ma, which agrees with the Falvey reconstruction. However, volcanics of the Tavua Volcanic Center on Viti Levu appear to have been rotated through even larger angles [*Malahoff, et al.*, 1982b], with paleomagnetic declinations

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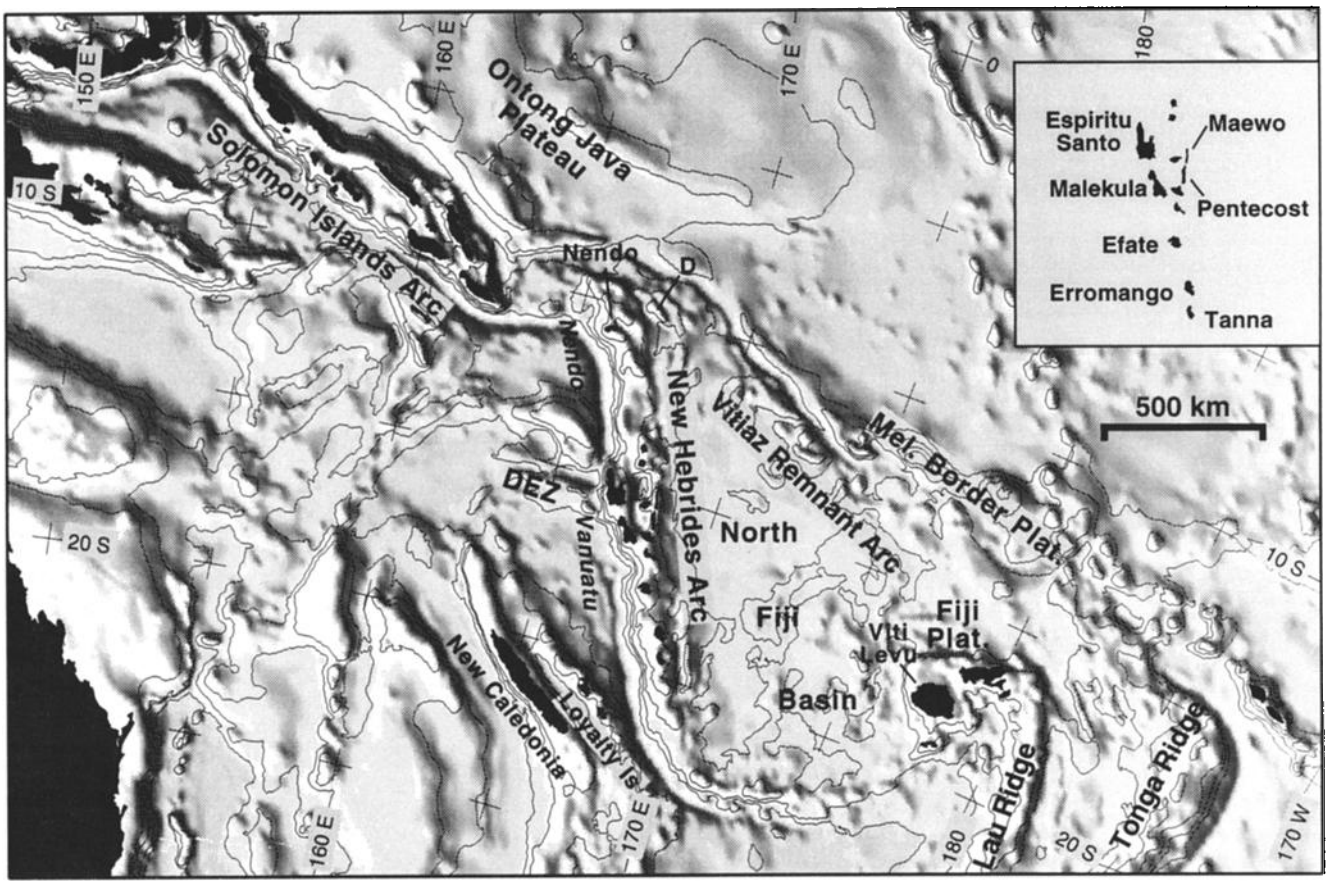


Figure 1. Regional tectonic setting of Nendo and the New Hebrides Arc. Bathymetry is indicated by sun-shaded ETOP05 data, overlain by 1000, 2000, 3000, and 4000 m contours. Projection is the same as used for the Circum-Pacific Map Project tectonic map, southwest quadrant [*Scheibner et al.*, 1991]: Lambert azimuthal equal-area, centered at 35°S, 135°E. Inset shows names of the major islands of Vanuatu. Text in italics indicates position of the informally defined "Vanuatu" and "Nendo" sectors of the New Hebrides Arc. DEZ, D'Entrecasteaux Zone aseismic ridge; D, Duff Group; Mel. Border Plat., Melanesian Border Plateau; Fiji Plat., Fiji Platform. Note intersection of linear trends of New Hebrides and Vitiaz Arcs at about 166°E, 9°S.

rotated through 70° for a rock with a K/Ar date of 7 Ma. Declinations of other undated volcanics of this association have been rotated by as much as 115° anticlockwise; *Malahoff et al.* [1982b] extrapolated their dated declination data to infer that rotation of Viti Levu began as early as 10 Ma.

A resolution of the conflict between the Falvey reconstruction and the evidence for greater rotation of the New Hebrides Arc may be that the paleomagnetic results from Vanuatu failed to detect the whole magnitude of the arc rotation. Both of the poles representing rocks older than 6 Ma in Falvey's study were drawn exclusively from formations on Malekula Island (Port Sandwich Formation and Tisman Volcanics). Anticlockwise rotation of Malekula through about 20° is a predicted consequence of a model of late Miocene-Holocene tectonism related to subduction of the D'Entrecasteaux Zone (DEZ) aseismic ridge [*Greene et al.*, 1988; *Mallick*, 1973], in which transverse wrench faulting in the central part of the arc is a response to the subduction of a seamount on the DEZ. If this rotation is removed from the two poles drawn from Malekula, the total rotation indicated for the New Hebrides Arc is similar to the rotation required for the reconstruction shown in Figure 3.

If Malekula fails to unambiguously record the full rotation of the New Hebrides Arc, other islands, away from the DEZ, could potentially do so. Nendo Island, politically part of the Solomon Islands and formerly known as Santa Cruz, lies near the northern end of the New Hebrides Arc, well away from the influence of the DEZ collision. Paleomagnetic analysis of arc rotation at Nendo also allowed a test of whether the arc has acted as a single microplate (with Malekula as a localized exception) or has moved as a series of discrete entities at different times, a long-standing question about the behavior of elongated arcs. Paleomagnetic sampling in Nendo offered an opportunity to test both the overall magnitude of New Hebrides Arc rotation and the relative timing of the rotation of the northern ("Nendo") and southern ("Vanuatu") segments of the arc.

2. Tectonic Setting

Back arc spreading in the North Fiji Basin since the late Miocene has followed the motion of the New Hebrides Arc southwestward, as the Australian Plate was subducted along the line of the North and South New Hebrides trenches. An interruption to the continuity of these trenches west of Espiritu Santo and Malekula Islands results from the subduction of the elevated DEZ [*Carney and Macfarlane*, 1982]. Spreading in the North Fiji Basin followed rifting of the earlier New Hebrides-

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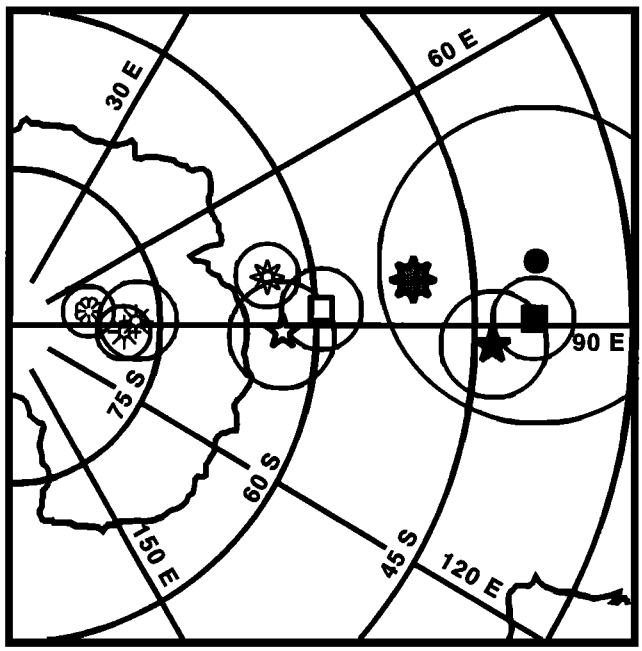


Figure 2. Southern Hemisphere projection of paleomagnetic poles from the New Hebrides Arc, with circles of 95% confidence. Vanuatu poles from *Falvey* [1978] indicating an apparent rotation of 28°. Open square and star are Falvey's poles for Malekula: solid symbols are equivalent poles after removal of inferred 20° anticlockwise local rotation of Malekula. Malue Beds pole is calculated as the mean of site poles for the reduced set of eight Malue Beds sites. Synthetic New Hebrides pole is calculated by rotating the south geographic pole around the New Hebrides-hotspot rotation pole.

Vitiaz Arc. Subduction below this arc formerly dipped southwestward, and the initiation of rifting and back arc spreading was accompanied by a reversal of arc polarity [*Karig and Mammerickx*, 1972; *Gill and Gorton*, 1973]. Polarity reversal in the Solomon Islands Arc has been seen as the result of "choking" of the former North Solomons Trench by collision with the thick crust of the Ontong Java Plateau [*Packham*, 1973]; by analogy, *Falvey* [1975] suggested that collision of the Melanesian Border Plateau with the Vitiaz Trench brought about reversal of the New Hebrides Arc.

Poles from Falvey's study dated to between 10 and 6 Ma formed a group with the 6 Ma pole, suggesting that rotation of the New Hebrides Arc had not begun until 6 Ma. Magnetic anomaly 4 has been identified over the oldest crust of the North Fiji Basin, however, indicating that spreading between the New Hebrides and Vitiaz arcs was already underway at about 8 Ma [*Malahoff et al.*, 1982a]. *Carney and Macfarlane* [1978] proposed that early North Fiji Basin lithosphere was formed prior to the arc polarity reversal by incipient back arc spreading behind and parallel to the Vitiaz Arc; rotation of the New Hebrides Arc was delayed until subduction shifted to the New Hebrides Trench.

Recent studies of the bathymetry and magnetic anomalies of the northwestern corner of the North Fiji Basin [*Pelletier et al.*, 1993] indicate a complex spreading history (Figure 5). An initial phase of spreading more or less parallel to the arc axis (phase 1) was followed by spreading on an east-west axis (phase 2). Phase 1 spreading, tentatively identified as anomalies 5A to 4, does not appear to extend into the extreme northwestern corner of the North Fiji Basin, west of 166.5°E. Phase 2 spreading (anomalies 3A to 2A) appears to represent the northwestward propagation of an east-west spreading system: latest phase 2 anomalies were identified as far west as 165.5°E, and the number of identifiable anomalies progressively decreases to the west in successive spreading segments. Phases 3 and 4 repeated the alternation of NE-SW and N-S spreading, but the axes of spreading were limited to the central and southern parts of the North Fiji Basin.

The Duff Group was rotated with the Vitiaz Arc in the reconstruction shown in Figure 3. Although it has been suggested that the Duff Group forms the western limit of the Vitiaz Arc [*Jezek et al.*, 1977], bathymetric and magnetic anomaly data [*Pelletier et al.*, 1993] indicate that the Duff Ridge on which the Duff Group lies is separated from both the Vitiaz Trench and the New Hebrides Arc by zones of back arc spreading. Structural [*Hughes et al.*, 1981] and paleomagnetic [*Musgrave*, 1986] evidence indicates that the islands of the Duff Group have been much more intensely deformed than neighboring Nendo, Utupua, and Vanikolo on the main axis of the New Hebrides Arc, suggesting that the Duff Ridge may be older and so may be a fragment of the axis of the Vitiaz Arc isolated by rifting on both sides.

Internal deformation of the arc resulting from collision with the DEZ is continuing today, as indicated by geodetic studies referring Global Positioning System (GPS) stations on New Caledonia and the Loyalty Islands, on the Australian Plate, to stations on the New Hebrides Arc [*Calmant et al.*, 1995; *Taylor et al.*, 1995]. The central section of the New Hebrides Arc, including the islands of Espiritu Santo, Malekula, Maewo, and Pentecost, is currently moving eastward relative to adjacent segments of the arc at about 35-85 mm/yr, apparently as a response to resistance to subduction of the elevated topography of the DEZ. The displacement is accommodated by crustal shortening on the eastern side of the arc and by strike-slip faults, indicated by their focal mechanisms, which crosscut the arc. The process may have been repeated at least twice, as individual seamounts on the DEZ were encountered [*Greene et al.*, 1988].

3. Geology of Nendo

The geology of Nendo was described in detail by *Craig* [1981], from which much of the following summary is drawn. Two major stratigraphic divisions can be recognized. The older sequence comprises the early Miocene Mengalu Volcanics and Nolua Volcanics. Disconformably above this volcanic sequence lies a suite of limestones and mixed carbonate-terrigenous sediments (represented in the study area by the Malue Beds) with

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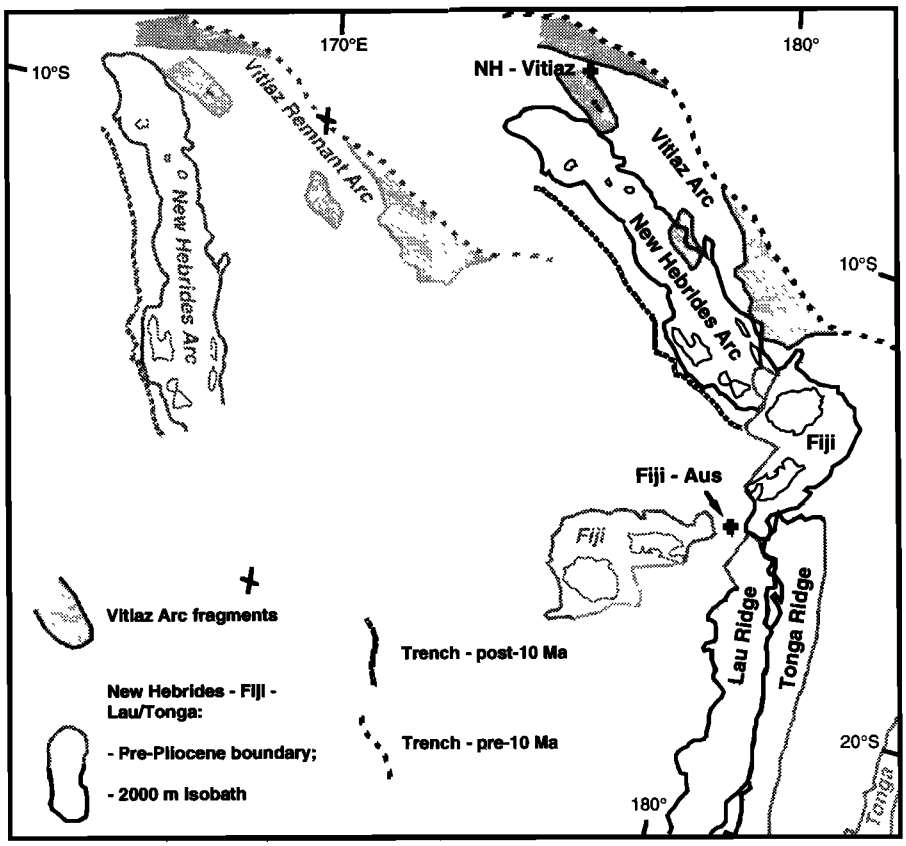


Figure 3. Reconstruction of the New Hebrides/Vitiaz-Fiji-Lau/Tonga arc system at 10 Ma, relative to a fixed Australian Plate. The projection is the same as that used in Figure 1. Boundaries of arc segments are based on the 2000 m isobath and the limits of pre-Pliocene crust, as indicated on the tectonic map of the Circum-Pacific region, southwest quadrant [*Scheibner et al.*, 1991]. Present-day positions of arc segments are shown in lighter density, with labels in italics; note that the Lau Ridge, which has remained fixed on the Australian Plate, does not move in the reconstruction. The rotation poles for the Fiji Platform relative to fixed Australia and the New Hebrides Arc relative to the Vitiaz Arc are indicated by crosses.

basaltic andesite lavas (the Mbo Lavas) intercalated near the top. The Malue Beds predominantly comprise well-bedded calcarenites, calcisiltites, and volcaniclastic silts deposited in a backreef lagoon. Benthic foraminifera are common and give a suggested age in the range from mid-Miocene to Pliocene. The stratigraphic thickness of the Malue Beds is at least 300 m.

Figurative and deformative bioturbation is pervasive in the Malue Beds [*Musgrave*, 1986]. Consequently, little of the depositional remanent magnetization (DRM) is likely to have been preserved. However, remagnetization by early postdepositional processes [*Verosub*, 1977; *Ellwood*, 1984a, b] is likely to have been acquired after bioturbation ceased and prior to dewatering of the sediment.

Renewed volcanism just prior to the deposition of the uppermost part of the Malue Beds produced the Mbo Lavas, which occur as a restricted strip of outcrop, representing less than 30 m of stratigraphy, of pillowed basaltic andesite flows and volcaniclastics. Folding is relatively gentle, and fold axes are horizontal to gently plunging, so the conventional paleomagnetic tilt correction can be used without risk of a significant declination error [*MacDonald*, 1980].

4. Analysis

4.1 Field Sampling

Site locations and stratigraphic positions are shown in Figure 6. Five sites were sampled in the Mbo Lavas (sites SI40-SI43 and SI47), and 21 sites were sampled in the Malue Beds (SI31-SI39, SI44-SI46 and SI48-SI56). Although the stratigraphic thickness of the sampled interval is difficult to determine, the sites appear to span a large proportion of the total thickness of the Malue Beds. Exposures of the Mengalu Volcanics are limited, and deep weathering makes them unsuitable for paleomagnetic sampling.

Each site was sampled by five (occasionally, six) cores. Cores were oriented by magnetic compass; weather conditions and rainforest cover usually precluded the use of a sun compass. However, magnetizations of all of the rocks sampled were not sufficiently high to produce any significant error in the magnetic compass. Cores in the Mbo Lavas were drilled in the central parts of pillows, away from selvedges. Cores were packed in sealed plastic bags to minimize their drying before they could be stored in field-free space.

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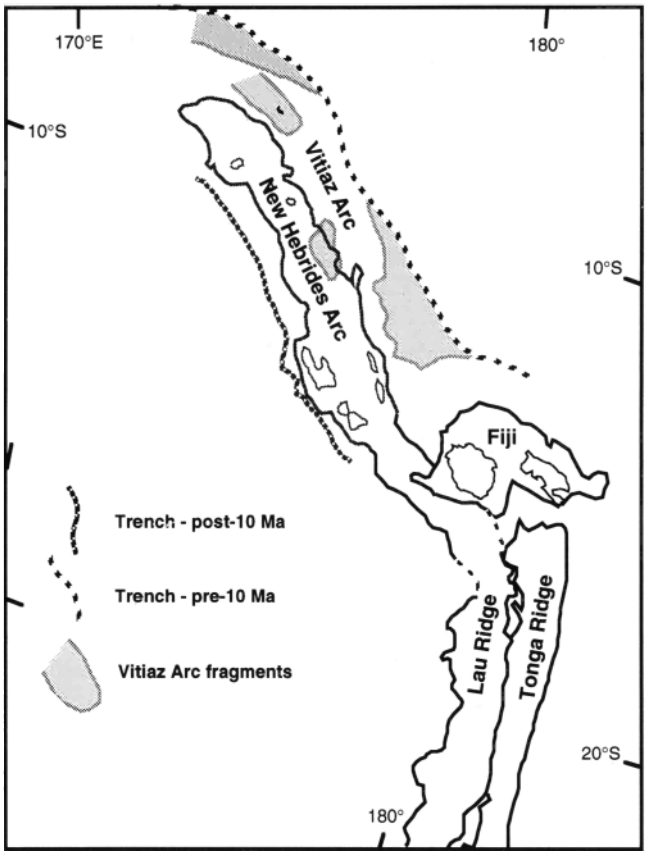


Figure 4. *Falvey's* [1978] reconstruction of the Outer Melanesian Arc system at 6 Ma, with the Australian Plate held fixed, on the same projection as Figures 1 and 3. Note the limited rotation of the Fiji Platform and its position offset northeastward from the main axis of the New Hebrides Arc. Unlike the new reconstruction shown in Figure 3, the southern part of the New Hebrides Arc, which is post-Pliocene, was retained in this reconstruction.

4.2. Paleomagnetic and Rock Magnetic Studies

4.2.1. Methods. Specimens were prepared by cutting cores to 2.5 cm lengths. Stepwise demagnetization was carried out on one specimen from each sample in the first instance, by alternating field (AF) means in the case of the Malue Beds and by thermal means for the Mbo Lavas [*Collinson*, 1983]. Demagnetization paths for the AF-cleaned sediments were frequently highly noisy, and duplicate specimens were demagnetized thermally. Magnetization was measured in most cases by a two-axis cryogenic magnetometer; where intensities exceeded the range of this device, a spinner magnetometer was used. Magnetometers and demagnetizing apparatus were magnetically shielded by Helmholtz coils. Specimens were stored in nearly field-free space throughout their demagnetization and measurement, although some exposure to laboratory fields and the Earth's field did inevitably occur.

Interpretation of demagnetization behavior was aided by the preparation of *Zijderveld* [1967] plots. Linear components of magnetization were fitted by principal component analysis [*Kirschvink*, 1980]. Hysteresis loops were determined on a set of AF-demagnetized specimens on a vibrating sample magnetometer at the University of California, Davis.

4.2.2. Malue Beds. Demagnetization behavior varied with stratigraphic position in the Malue Beds (Figure 7). Sites SI50-SI56, in the lower part of the Malue Beds, all reveal the presence of a linear component convergent on the origin of the Zijderveld plot (the "characteristic magnetization," likely to be a primary, early postdepositional remanence) under both AF and thermal demagnetization. Alternating fields of 20-30 mT are sufficient to remove a normal overprint with a direction (without correction for dip) close to that of the dipole field, implying that it was acquired in situ at some time following tilting of the beds. The characteristic remanence of most sites is reversed; a single site (SI53), near the base of the Malue Beds, has a normally polarized characteristic remanence.

More distinct differences between AF and thermal demagnetization paths are apparent in specimens from sites SI44-46 and SI48-49, which were sampled in a cluster near the stratigraphic level of the Mbo Lavas. Noise is present in the AF demagnetization paths at applied fields above about 30 mT. A normally polarized overprint is again present; however, this cannot be isolated from the characteristic magnetization by AF demagnetization. Thermal demagnetization results in much clearer separation of the stability spectra of the overprint and characteristic magnetization; a linear, convergent, characteristic magnetization can be isolated for all but one site. The polarity of the characteristic remanence is reversed throughout this part of the sequence.

Sites near the top of the sampled sequence (SI31-39) display a wide variety of responses to both AF and thermal demagnetization. Although a reversed characteristic magnetization overprinted by a normal component is revealed in a few specimens by AF demagnetization and in a few of the duplicate specimens by thermal demagnetization, many specimens only show evidence of a normal component. However,

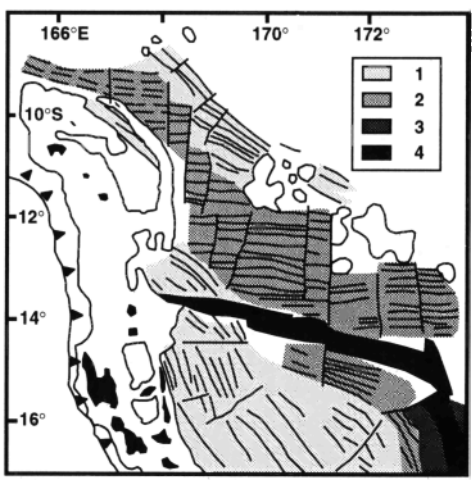


Figure 5. Spreading in the northwestern corner of the North Fiji Basin [after *Pelletier et al.*, 1993]. Trends of anomalies and fracture zones are indicated. Individual anomalies were not identified, so absolute ages are not known. Instead, four phases of spreading were recognized, with phase 1 being the oldest and phase 4 the youngest; these are indicated in the figure by the four shaded zones.

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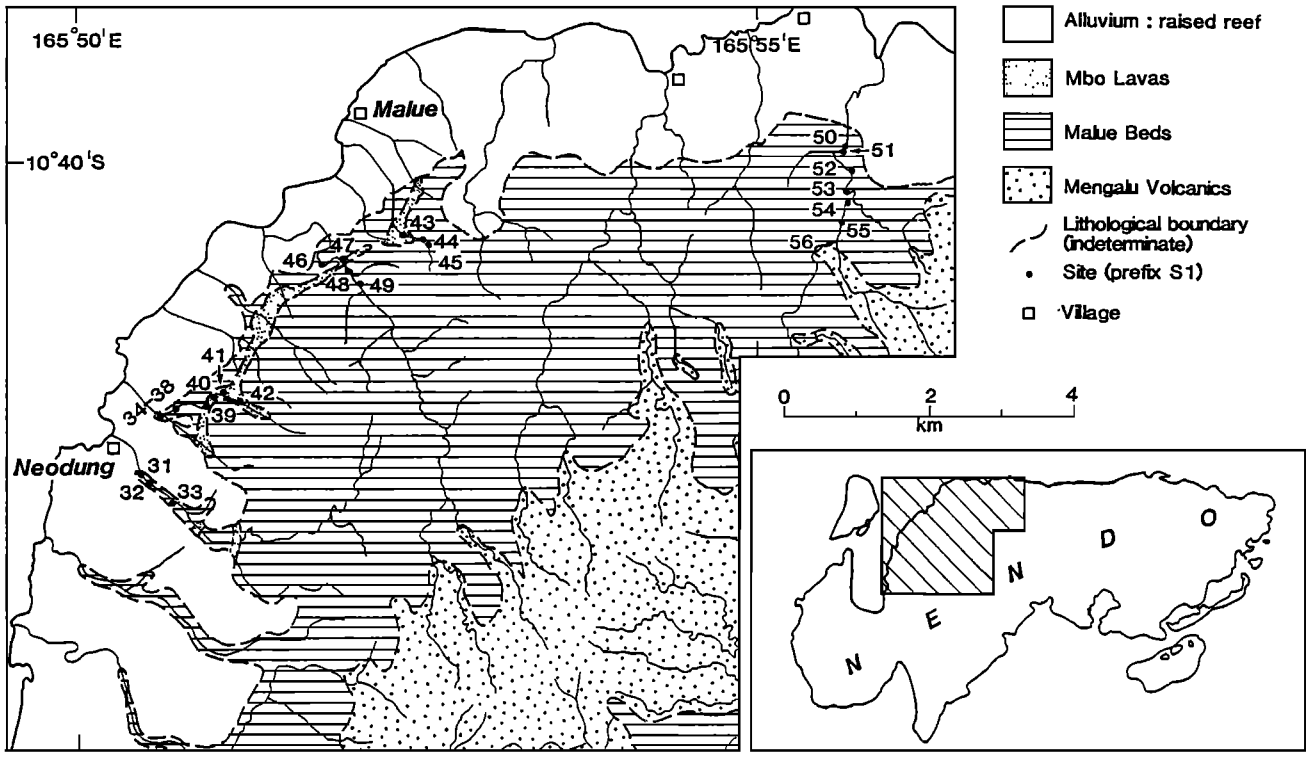


Figure 6. Geology of northwestern Nendo and site locations. The Malue Beds mostly dip gently northwest to north, away from the underlying Mengalu Volcanics; the Mbo Lavas are intercalated near the top of the Malue Beds.

this normal component is consistently close to the axial geocentric dipole field direction prior to tilt correction. The normal component is evidently a postdepositional overprint, similar to that affecting sites lower in the sequence, but with a stability spectrum extending to high coercivities and in some specimens to demagnetization temperatures close to the magnetite Curie point. In all specimens in this set of sites where a distinct inflection in the demagnetization path can be recognized, allowing the polarity of the characteristic component to be determined, the characteristic component is reversed. Hence it is probable that the characteristic remanence of all of these sites is reversed.

Specimens from the Malue Beds all exhibit relatively simple hysteresis loops which reach saturation by about 0.3 T (Figure 8), without any evidence of "wasp-waisted" behavior [*Channell and McCabe*, 1994], indicating that high-coercivity phases (e.g., hematite) are not present in significant proportions. All thermally demagnetized specimens are demagnetized by 590°C, consistent with the presence of magnetite. Thermal demagnetization curves do not show any clear inflection at around 300°C, as would be expected if magnetic sulfides (pyrrhotite and/or greigite) or maghemite were present in significant quantities. All hysteresis specimens have a saturation remanence (M_{rs}) to saturation magnetization (M_s) ratio between 0.05 and 0.2, in the lower range of the pseudo-single domain (PSD) field [*Day et al.*, 1977], and distinct from the low-coercivity superparamagnetic/multidomain assemblage characteristic of remagnetized carbonates [*Channell and McCabe*, 1994]. The M_{rs}/M_s ratio does not show any systematic relationship to stratigraphic position. Coercive force (H_c) does vary systematically with stratigraphic position, however. Specimens from sites SI31-SI39, at the top of the sequence, all have $H_c \leq 5$ mT; specimens from the rest of the sequence have $H_c > 5$ mT.

Overprinting in specimens from sites SI31-SI39 is associated with a decrease in H_c. Lower coercivity may have made these specimens more susceptible to remagnetization by viscous remanent magnetization (VRM), which would be expected to have a direction close to the dipole field direction (if it has been accumulated over a long timescale). However, the high AF fields needed to remove this component are inconsistent with VRM. Alternatively, the overprint may be a chemical remanence, acquired by the growth of magnetite after the beds were tilted, possibly by the oxidation of iron sulfides during weathering. Similarly intense and stable overprinting (often interpreted, rightly or wrongly, as a VRM) has frequently been reported from carbonates and silts in carbonate associations [e.g., *Creer*, 1959; *Achache et al.*, 1982; *Scotese et al.*, 1982].

4.2.3. Mbo Lavas. Specimens from sites SI40-SI42 from the Mbo Lavas are noteworthy for the presence of a partial self-reversal. Each shows a normally polarized, lower stability component which is closely antiparallel to the characteristic component (Figure 9). Although a normal component could arise by VRM or by thermal or chemical overprinting, the case for self-reversal is strengthened by the observation that this antiparallelism persists from specimen to specimen even though the direction of the characteristic magnetization varies between pillow extrusions. A slight divergence between the characteristic direction and the self-reversal overprint results from a further overprint, probably VRM, imposed over both. Some mineralogical evidence exists to support the observation of partial self-reversal in the Mbo Lavas. *Craig* [1981] noted the presence of skeletal growths of ilmenite on euhedral magnetite grains in

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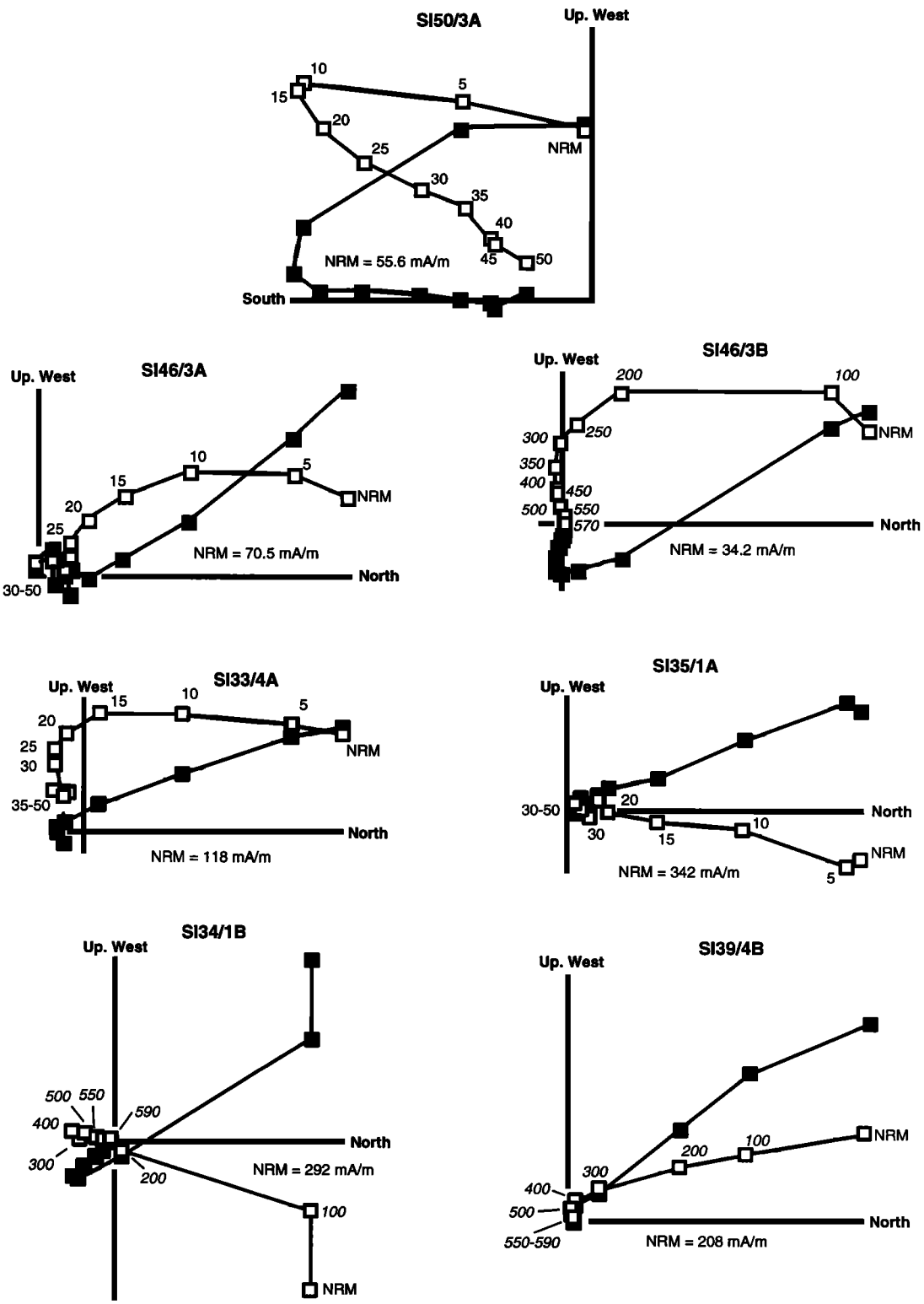


Figure 7. Zijderveld plots of AF and thermal demagnetization paths showing the range of demagnetization behavior of specimens from the Malue Beds. Open squares show the horizontal (declination) projection of demagnetized remanences; solid squares show the projection on the north-south vertical plane. AF demagnetization intensity (in mT) is indicated by plain text figures; thermal demagnetization (in °C) is indicated by italic text. NRM intensity for each specimen is indicated. Specimen SI50/3A was sampled near the base of the Malue Beds. A reversed characteristic magnetization (ChRM) and normal overprint can be easily identified under AF demagnetization. Specimens SI46/3A and SI46/3B were sampled from the same core, drilled near the stratigraphic level of the Mbo Lavas. Note how thermal demagnetization reveals the presence of reversed ChRM and normal overprint, which are not separated by AF demagnetization. Specimens SI33/4A, SI34/1B, SI35/1A, and SI39/4B were sampled near the top of the Malue Beds. AF demagnetization of specimen SI33/4A reveals both the reversed ChRM and normal overprint; similar demagnetization of specimen SI35/1A indicates only a single component, which appears to be the overprint. Thermal demagnetization of SI34/1B clearly reveals both the reversed ChRM and normal overprint; similar demagnetization of SI39/4B suggests the presence of a weak, reversely polarized component imperfectly isolated above 550°C.

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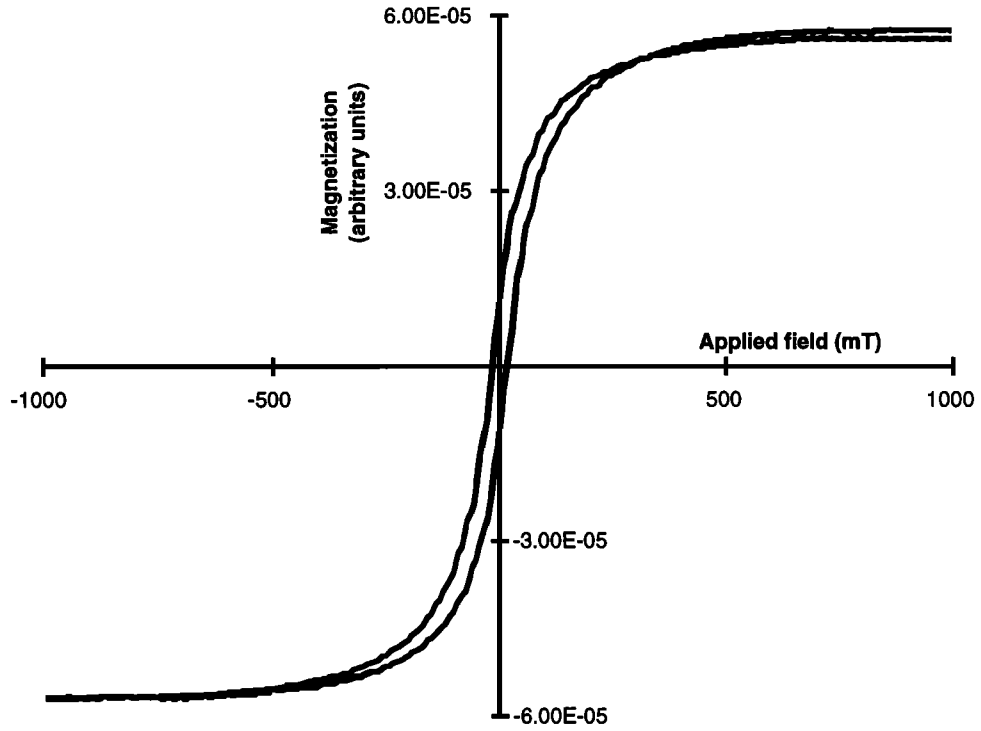


Figure 8. Hysteresis loop for Malue Beds specimen SI44/4A. The curve suggests remanence is carried by magnetite, without significant contribution from antiferromagnetic phases (hematite or pyrrhotite).

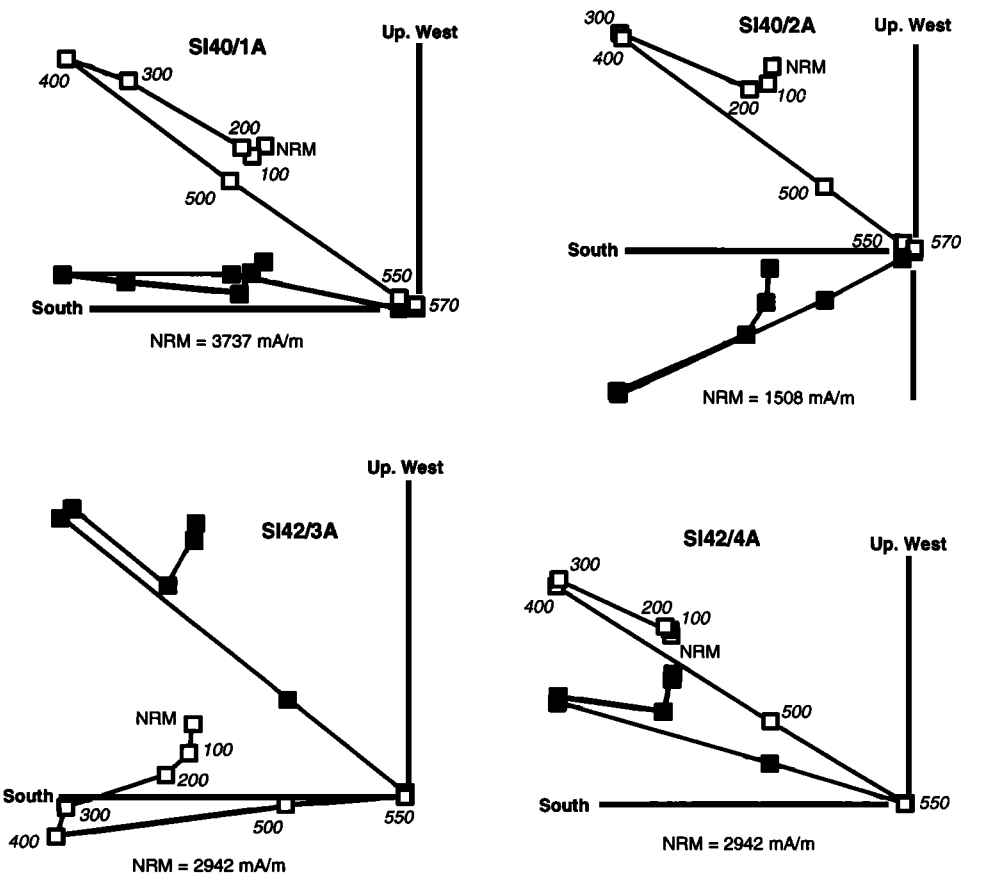


Figure 9. Thermal demagnetization of representative specimens from the Mbo Lavas revealing a partial self-reversal. The self-reversed component, present up to 400°C, always remains antiparallel to the primary component, despite variations in the primary direction. Viscous overprinting slightly offsets the antiparallelism. Plots and symbols are as for Figure 7.

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Table 1. Characteristic Magnetizations from Nendo

Site	Stable?	Polarity	Uncorrected		Bedding Corrected		N	k	α_{95} deg
			D deg	I deg	D deg	I deg			
					Malue Beds				
SI31*	yes	R	208.8	12.2	210.8	5.0	3	25.0	25.2
SI32	no								
SI33*	yes	R	186.4	47.5	212.3	43.7	4	7.2	36.8
SI34	yes?†	R?							
SI35	no								
SI36	no								
SI37	no								
SI38	no								
SI39	no								
SI44	yes	R	256.4	7.8	255.4	6.8	3	133.8	10.7
SI45	no								
SI46	yes	R	260.8	37.0	259.4	30.9	3	180.6	9.2
SI48	yes	R	231.7	40.2	232.5	25.8	5	22.5	16.5
SI49	yes	R	248.4	55.3	254.8	31.1	5	187.7	5 6
SI50	yes	R	212.1	-0.5	212.5	16.8	6	90.0	7.1
SI51	yes	R	204.3	28.1	202.2	28.7	5	43.7	11.7
SI52*	yes	R	245.1	15.0	247.0	13.8	4	22.1	20.0
SI53‡	yes	N	50.8	-21.2	48.4	-25.6	5	18.2	18.4
SI54*	yes	I	105.2	2.0	108.8	-18.6	5	23.5	16.1
SI55	yes	R	214.6	7.5	213.9	17.6	5	24.2	15.9
SI56*	yes	I/R	256.7	-34.0	251.0	-2.8	6	17.6	16.4
					Mbo Lavas§				
SI40	yes	R	210.4	8.8			5	44.5	11.6
SI41	yes	R	190.2	10.4			6	19.9	15.4
SI42	yes	R	204.3	-23.7			5	22.2	16.6
SI43	no								
SI47	no								

For Malue Beds, for mean corrected direction, declination $D = 232.2°$, Inclination $I = 24.3°$, $N = 8$, $k = 13.9$, $\alpha_{95} = 15.4°$; and for corresponding mean pole, longitude is = 084.0°E, latitude is 38.8°S, $N = 8$, $k = 13.7$, $\alpha_{95} = 15.5°$

*Omitted from calculation of Malue Beds mean.

†Polarity identified, but component direction could not be isolated from overprints.

‡Direction inverted in calculation of Malue Beds mean.

§Structural correction not known, but dip probably <30°.

thin-sections of the Mbo Lavas. *Heller and Petersen* [1982] observed both partial and complete self-reversal affecting lavas in which low temperature oxidation of titanomagnetite had resulted in intergrowths of ilmenohematite in a pattern similar to that of the Mbo Lavas.

Apart from this behavior, the path of the demagnetization vector in sites SI40-SI42 is relatively simple. All specimens converge linearly on the origin of the Zijderveld plot at demagnetization temperatures above 500°C. The Curie point is less than 570°C, and sometimes less than 550°C, suggesting that a titanomagnetite, with a varying proportion of Ti, is the dominant magnetic carrier. The characteristic component is uniformly reversed in polarity. Sites SI43 and SI47 are inconsistent from specimen to specimen. The primary magnetization has been entirely replaced in these two sites, perhaps by weathering or lightning-induced components.

4.2.4. Interpretation. Stable characteristic magnetizations could be isolated from 13 sites in the Malue Beds and three sites in the Mbo Lavas (Table 1). With the exception of sites SI53 and SI54, which appear to record a short interval of normal polarity and an intermediate polarity transition, the characteristic magnetization of all sites which have preserved a stable characteristic magnetization is reversely polarized. It is not possible to provide an unequivocal, independent test of whether the characteristic magnetization is primary in either the Malue Beds or the Mbo Lavas. Dips in the Malue Beds are mostly less than 20°, restricting the power of any form of the fold test [*McElhinny*, 1964; *McFadden and Jones*, 1981]. Eleven sites (excluding intermediately polarized SI54 and the outlying site SI56) define the characteristic remanence of the Malue Beds. Using the method of *Watson and Enkin* [1993], which examines concentrations of directions at varying degrees of untilting, the maximum concentration of site mean directions from these 11 sites occurs at 139% ± 20% untilting. The excess beyond 100% untilting may reflect the large confidence limits of three of these sites (SI31, SI33, and SI52), which have $\alpha_{95} \geq 20°$. Recalculation omitting these three sites gives a maximum concentration at 124% ± 20% untilting, indicating that the characteristic remanence can be interpreted as predating folding. As the mean direction of the reduced set of sites (declination $D = 232.2°$, inclination $I = 24.3°$, $N = 8$, $k = 13.9$, $\alpha_{95} = 15.4°$) is not significantly different from the mean including the sites with large α_{95} ($D = 230.0°$, $I = 23.6°$, $N = 11$, $k = 13.2$, $\alpha_{95} = 13.0°$), the reduced set has been used to calculate the formation mean direction and pole.

Bedding could not be determined for the Mbo Lavas sites, so no fold test was available for this unit. However, in the absence of any obvious source for a uniformly reversed overprint on both Malue Beds and Mbo Lavas, the characteristic magnetization of both can be reasonably assumed to be primary.

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With the exception of the intermediately polarized site SI54, the characteristic magnetization of all sites in the Malue Beds has been rotated clockwise. The mean direction indicates a total clockwise rotation from geographic north of 52.2° ± 12.1° (95% confidence range calculated using the method of *Debiche and Watson* [1995]). There has been little change in latitude; the paleolatitude equivalent to the inclination of the mean direction is 12.7°, with a 95% confidence range from 7.1° to 19.5° (calculated by the Debiche and Watson method for the inclination), while Nendo's present latitude is 10.8°. No significant "inclination error" (resulting in a shallowing of the recorded inclination) has occurred, either through rotation of particles during deposition [*King*, 1955; *Verosub*, 1977] or by compaction-induced rotation of postdepositional remanence [*Anson and Kodama*, 1987; *Arason and Levi*, 1990; *Deamer and Kodama*, 1990; *Levi and Banerjee*, 1990]. Averaging site mean poles yields a mean pole for the Malue Beds at 38.8°S, 84.0°E with a 95% confidence radius of 15.5°.

As local structural control for the Mbo Lavas is lacking, no reliable quantitative measure of rotation can be gained from this formation. However, they confirm the pattern of clockwise rotation and of reversed polarity.

4.3. Calcareous Nannofossil Biostratigraphy

Samples from the Malue Beds were examined for calcareous nannofossils. Ages for nannofossil biostratigraphic events are based on the compilation by *Young et al.* [1994], correlated to the revised magnetostratigraphic timescale of *Cande and Kent* [1995]. Samples from sites SI56 to SI46 contain rare *Pseudoemiliania lacunosa*, few *Reticulofenestra pseudoumbilica*, and abundant *Sphenolithus abies/neoabies*. Based on the overlap of these three taxa, sites SI56 to SI46 are assigned to Zone CN11, of late early Pliocene age. Other taxa present which support an age of no older than early Pliocene are *Oolithotus antillarus*, *Ceratolithus armatus*, and *Ceratolithus cristatus*. The presence of *Discoaster surculus* in the sample from site SI33, the stratigraphically highest sample, indicates an age of no younger than late Pliocene (Subzone CN12b), assuming that *D. surculus* is not reworked.

4.4. Magnetostratigraphy

Only a single interval of normal polarity, restricted to the interval represented by sites SI53 and SI54, is detected in the three stratigraphic sections sampled, which appear to represent a substantial part of the Malue Beds including both their base and top. The interval from Zone CN11 to Subzone CN12b includes mixed normal and reversed polarity intervals of the Gilbert, Gauss and Matuyama chrons (Figure 10): however, Zone CN11 lies entirely in the reversed Gilbert Chron, and the first occurrence of *Pseudoemiliania lacunosa* lies just below the top of the normal Cochiti Subchron. If sites SI53 and SI54 lie a little below site SI56 (likely, given the limited stratigraphic control available in the field), then the normal polarity interval represented by sites SI53 and SI54 is probably the Cochiti Subchron. The uniform reversed polarity of the rest of the sites suggests that the whole sequence lies within the upper part of the Gilbert Chron. In the *Cande and Kent* [1995] timescale, this restricts the age of the sequence to between about 4.29 Ma (base of Cochiti Subchron) and 3.58 Ma (top of Gilbert Chron).

The restricted magnetostratigraphic and biostratigraphic age range for the Malue Beds indicates deposition of the sampled interval (and probably of the whole sequence, estimated to be at least 300 m thick) in less than a million years. High rates of deposition are consistent with the back-reef lagoonal facies envisaged for this unit [*Craig*, 1981].

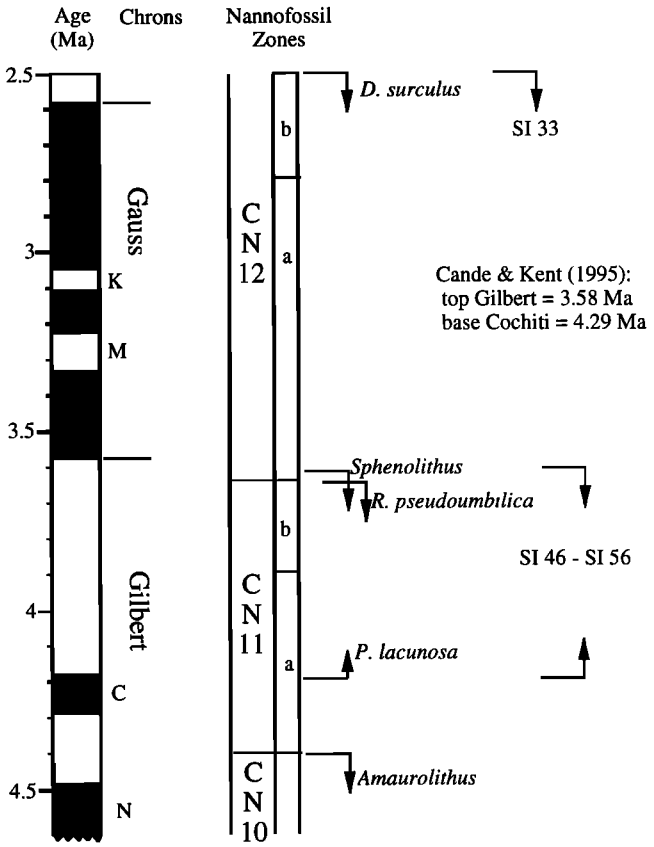


Figure 10. Nannofossil constraints on ages of the Malue Beds sequence and a reference magnetostratigraphy. Arrows on the right bracket age limits for the sites indicated; given these, and the uniformly reversed polarity of all but the bottom of the sequence, the Malue Beds sequence must lie in the upper Gilbert Chron.

5. Discussion

5.1. Rotation of the Nendo Sector and Propagation of North Fiji Basin Spreading

Nendo has recorded far more clockwise rotation than was detected in the *Falvey* [1978] study of Vanuatu (Figure 2). Evidence of internal deformation within Nendo (intense folding or the development of shear zones) is lacking, so it appears unlikely that this additional rotation reflects a local tectonic process restricted to within Nendo. The simplest interpretation is that the rotation of Nendo reflects the motion of the northern part ("Nendo sector") of the New Hebrides Arc (although the example of Malekula does illustrate the possibility of a more complex, localized rotation history).

Despite the fact that the rotation of Nendo recorded in the Malue Beds has occurred later than 4.29 Ma, this rotation is comparable with the total rotation needed to reconstruct the New Hebrides Arc to its presumed late Miocene position, as shown in Figure 3. The agreement between the paleomagnetic and geometric reconstruction rotations can be tested by comparison of the Nendo pole with a synthetic paleomagnetic pole for the New Hebrides Arc. If it is assumed that subduction along the Vitiaz Trench had completely ceased prior to rotation of the New Hebrides Arc and was not reinitiated at any time during the

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spreading of the North Fiji Basin, the Vitiaz Arc can be taken to be fixed in the Pacific Plate during opening of the North Fiji Basin. Consequently, a synthetic paleomagnetic pole for the New Hebrides Arc can be obtained by rotating the south geographic pole around the composition of the New Hebrides-Vitiaz and Pacific-hotspot rotation poles, and the reconstruction of the New Hebrides Arc relative to a fixed Australian Plate (as shown in Figure 3) involves composition of the New Hebrides-Vitiaz and Pacific-Australia rotation poles. Note that this differs from the assumptions involved in the reconstruction by *Hamburger and Isacks* [1987], who rotated the New Hebrides Arc around a pole fixed in the Australian Plate while holding the Vitiaz Arc on the Pacific Plate, which resulted in a 400-500 km gap between the New Hebrides Arc and the Vitiaz Trench in their late Miocene reconstruction (Hamburger and Isacks recognized this problem and ascribed it to unconstrained translational motion of the New Hebrides Arc; a translation, however, merely represents a rotation around a distant pole, and the combination of rotation and translation they described is more simply defined by a rotation around a pole fixed in the Pacific Plate). Pacific Plate absolute motion changed significantly between about 5 and 3 Ma [*Cox and Engebretson*, 1985]. Composing the New Hebrides-Vitiaz reconstruction pole and the *Pollitz* [1986] Pacific-hotspot pole for 3.2 Ma gives a net absolute motion pole for the New Hebrides Arc at latitude 4.9°S, longitude 167.8°E, rotation 38.5° anticlockwise. The corresponding synthetic paleomagnetic pole position is latitude 51.6°S, longitude 79.5°E, which lies within the confidence limits of the Malue Beds paleomagnetic pole (Figure 2).

Although the rotations of the Nendo and Vanuatu sectors of the arc appear to have a similar magnitude, they have clearly occurred at different times. The position of the 6 Ma pole from the Vanuatu sector (from volcanics on Pentecost and Maewo, unaffected by the rotation of Malekula) indicates that rotation of the Vanuatu sector of the arc had already commenced between 10 and 6 Ma. It appears that rotation of the Nendo sector "caught up" with the rotation that had already occurred in the Vanuatu sector to produce the nearly collinear system seen today. Motion of different sectors of the arc may in fact have alternated episodically, and this may correspond to the series of phases of location and direction of spreading seen in the North Fiji Basin. The tendency for these individual phases of motion of different arc segments to sum to an approximately linear trend for the arc overall may result from a mechanism constrained by minimization of the deformation of the downgoing plate to avoid propagation of tear faults in the Australian Plate between arc segments. As the downgoing plate in front of a lagging segment bends to attempt to follow its adjoining part in front of an advancing segment of the arc, the resulting trench rollback will prompt the lagging arc segment to advance and catch up with the rest of the arc.

Propagation of phase 2 spreading into the northwestern corner of the North Fiji Basin would have facilitated rotation of the Nendo segment of the arc; phase 2 spreading is therefore younger than 4.29 Ma. This reorganization of the local tectonic system may have been stimulated by the early Pliocene change in Pacific Plate motion.

5.2. Reconstruction of the New Hebrides Arc and Fiji Platform

With the New Hebrides Arc restored to its former position relative to the remnant Vitiaz Arc as part of the Pacific Plate, the two arcs can be restored to their position relative to a fixed Australian Plate (which includes the Lau Ridge) at 10 Ma (Figure 3), by rotation around the Pacific-Australia reconstruction pole derived from *Stock and Molnar* [1982]. Only the pre-late Miocene part of the New Hebrides Arc has been restored; the arc south of Malekula and Maewo is regarded as a construction built during late Miocene to present subduction along the New Hebrides Trench. The gap remaining between the Miocene New Hebrides-Vitiaz Arc and the Lau Ridge is sufficient to accommodate the Fiji Platform; Figure 3 shows a plausible position of the platform, involving a rotation of Fiji relative to the Australian Plate of 140° around a pole at 16.5°S, 179.5°W. Of course, other rotations of the Fiji Platform would be consistent with the reconstructed positions of the other arcs, but two points are clear; rotations of the Fiji Platform ≥ 115°, as postulated by *Malahoff et al.* [1982b], are clearly possible, and the Fiji Platform can form part of a continuous chain with the New Hebrides Arc and Lau Ridge.

6. Conclusions

Biostratigraphy and magnetostratigraphy date the rotation of the Malue Beds paleomagnetic pole to no earlier than 4.29 Ma. Since this time, Nendo (and, presumably, the Nendo sector) has recorded the same total rotation as the Vanuatu sector of the New Hebrides Arc has experienced since about 10 Ma. Rotation in the Vanuatu sector commenced earlier: differences in timing of rotation between the Nendo and Vanuatu sectors reflect phases of spreading in different parts of the North Fiji Basin. Propagation of phase 2 spreading in the North Fiji Basin occurred after 4.29 Ma and may have been a consequence of the Pliocene change in Pacific Plate motion. The apparently similar magnitude of the rotation of the two sectors of the arc, despite differences in timing, suggests that minimization of deformation of the downgoing plate tends to keep sectors of the arc aligned.

The magnitude of the rotation indicated by the Nendo paleomagnetism supports the suggestion that the *Falvey* [1978] paleomagnetic study underestimated the overall rotation of the New Hebrides Arc. The reconstruction of the New Hebrides and Vitiaz Arcs and the Fiji Platform based on the earlier paleomagnetic study was constrained by the smaller rotation. Removal of this restriction, and deletion of the pre-Pliocene parts of the New Hebrides Arc, allows a more tectonically satisfactory reconstruction, in which the Fiji Platform originally lay on the main axis of the former contiguous arc system. The new reconstruction allows for a much larger rotation of Fiji than did the earlier reconstruction, in line with paleomagnetic evidence for at least 115° of rotation for Fiji.

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